MANCHESTER 1824
The University of Manchester

EART11100

How to Build a Planet

Week 7

The Archean Earth: *4000-2500 Ma*

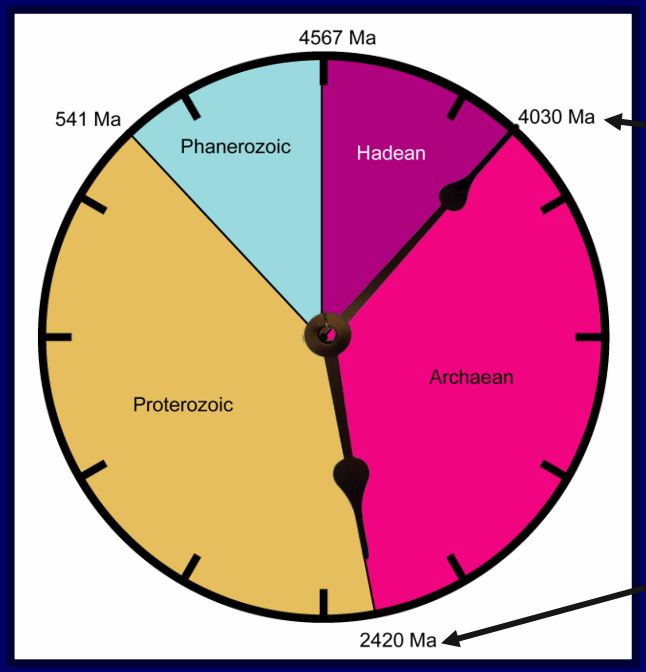
'That dark backward and abysm of time' (*William Shakespeare*)

The birth of the continents and life

Key events in the Archean

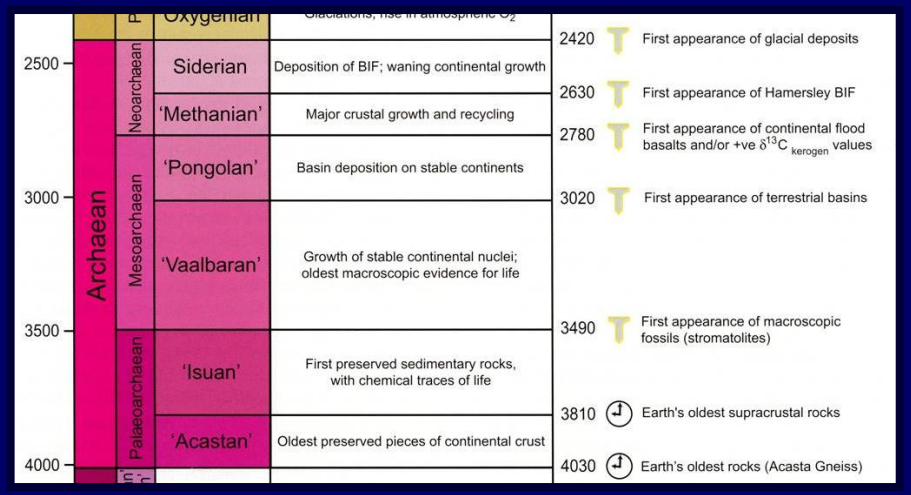
- The rock record begins at 4030 Ma
- Most continental crust formed (60-70% current volume)
- The first continents stabilised
- The tectonic style of the Earth evolved from lid tectonics to early form of plate tectonics
- Thick sequences of sediments were deposited in the deep oceans and on shallow continental shelves
- The first life developed (microbial).
- Enigmatic iron-rich sedimentary rocks formed that have not formed since.

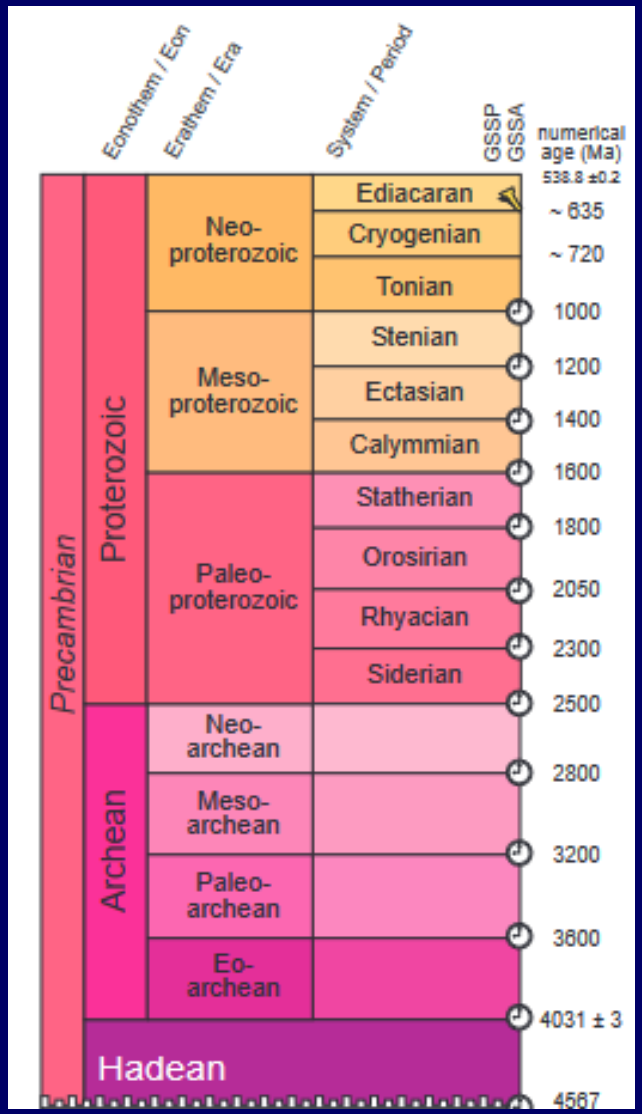
The Archean Earth: Greek ‘beginning’ or ‘origin’



oldest preserved rocks on Earth

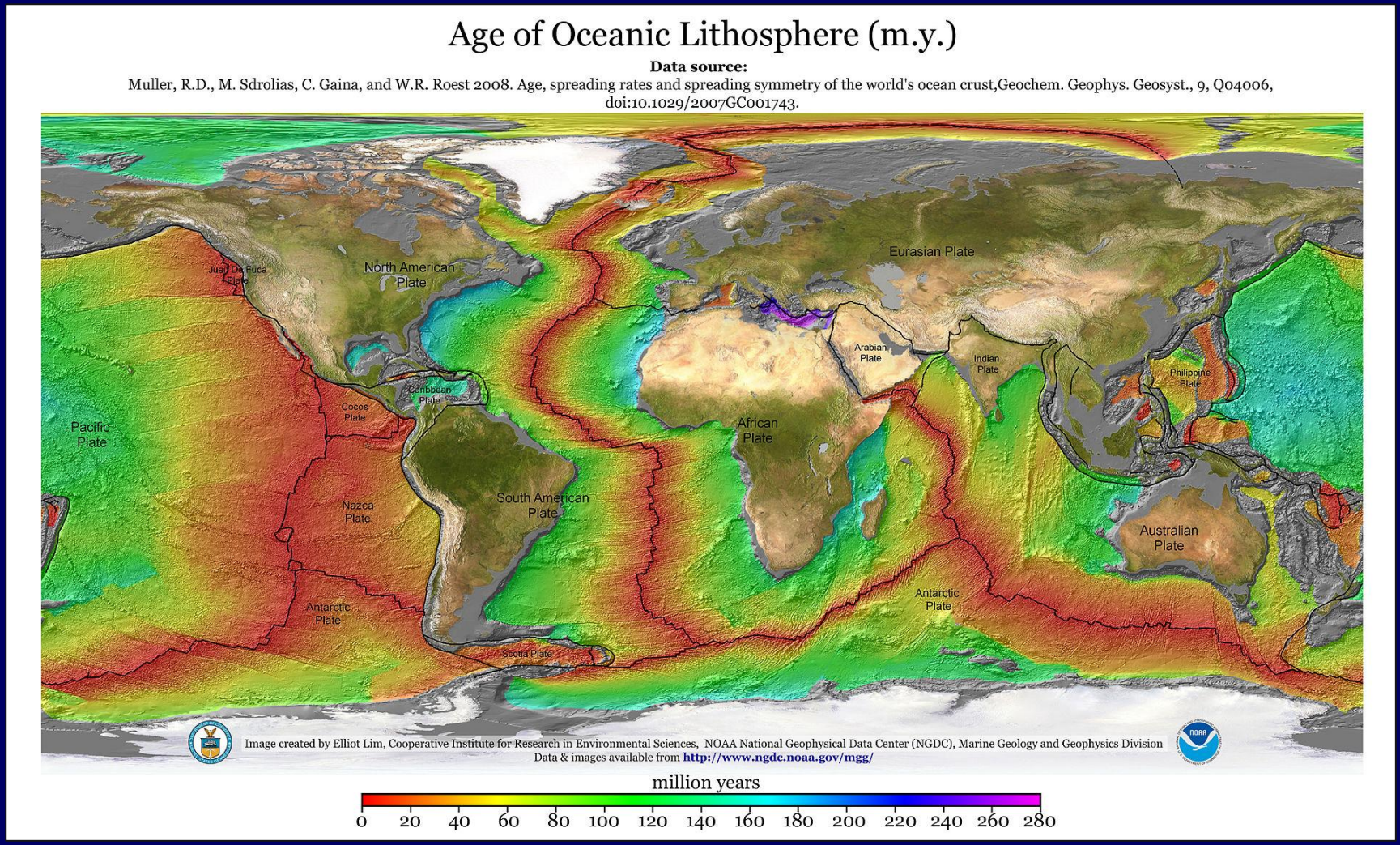
first glacial deposits





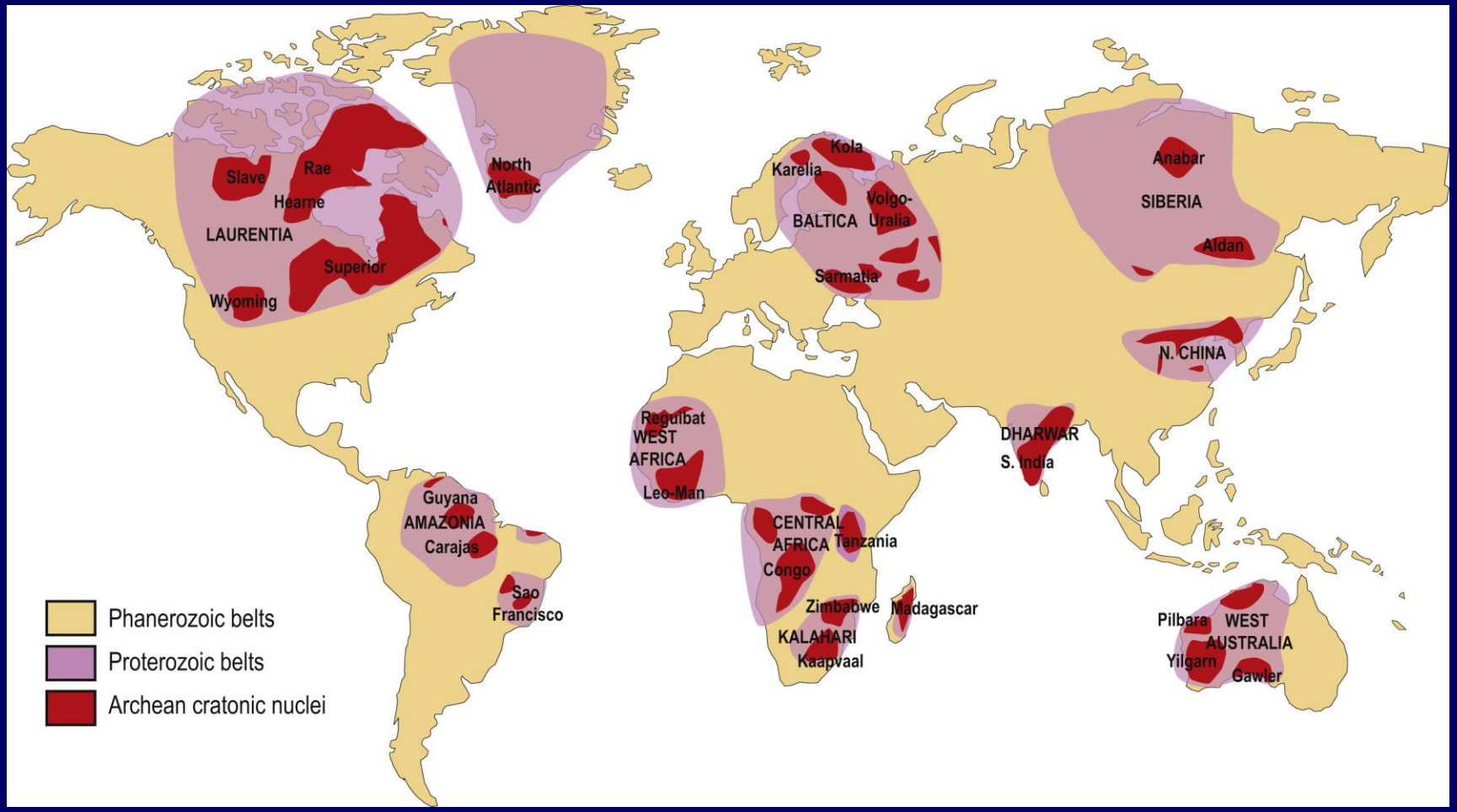
Note spelling: **Archaean** **Palaeo**

Where to find the oldest rocks on the Earth today



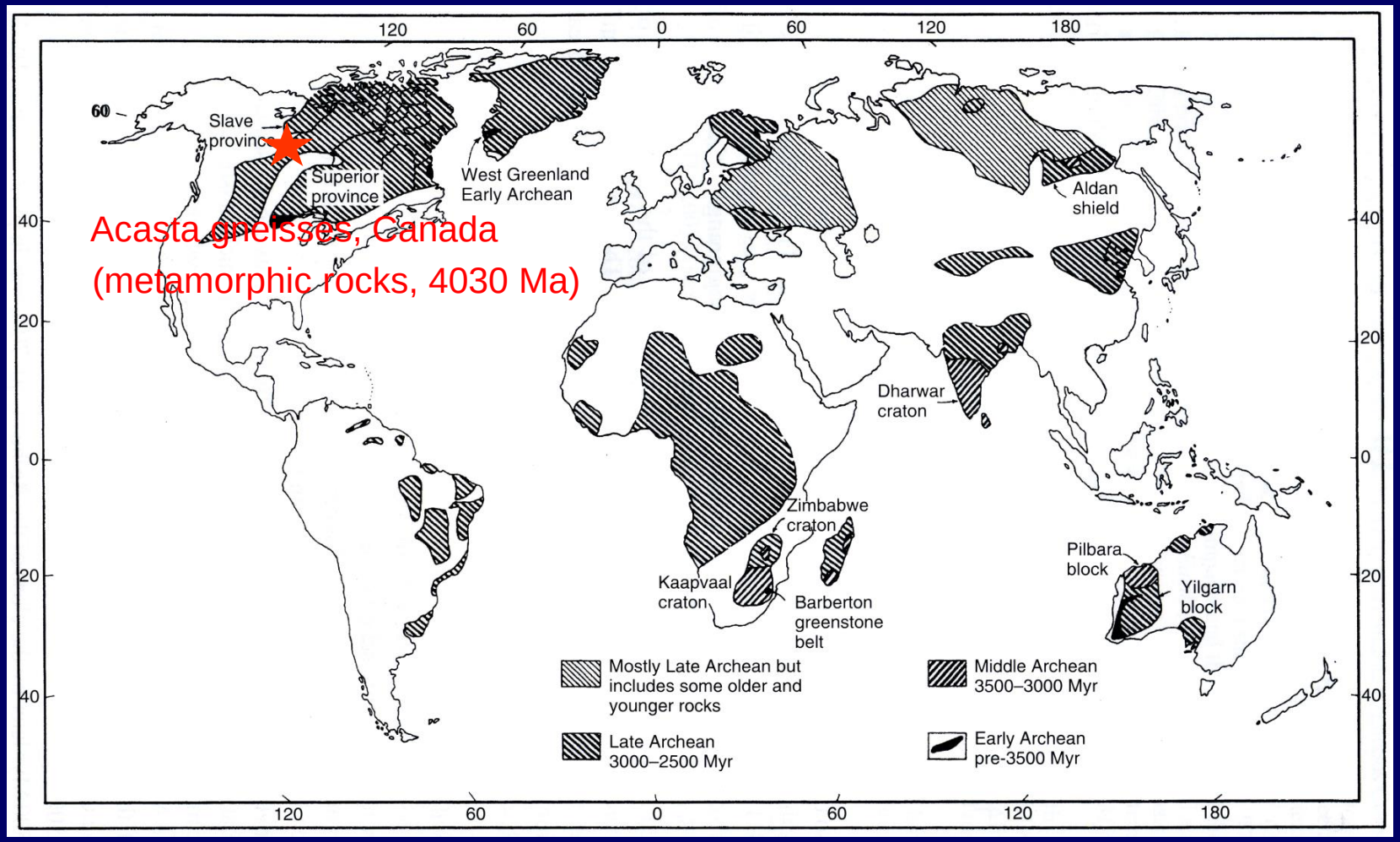
The oldest oceanic crust in the oceans today are <300Ma

Age of continental cratons around the world today



- Archean age rocks form the nuclei of every continental craton (35-40 nuclei)
- Craton (Greek for ‘strength’): large, tectonically stable regions of the continental lithosphere

Distribution of Early, Mid and Late Archean cratons



Very few cratonic areas of the Earth are older than >3500 Ma

Defining the base of the Archean- the oldest rocks



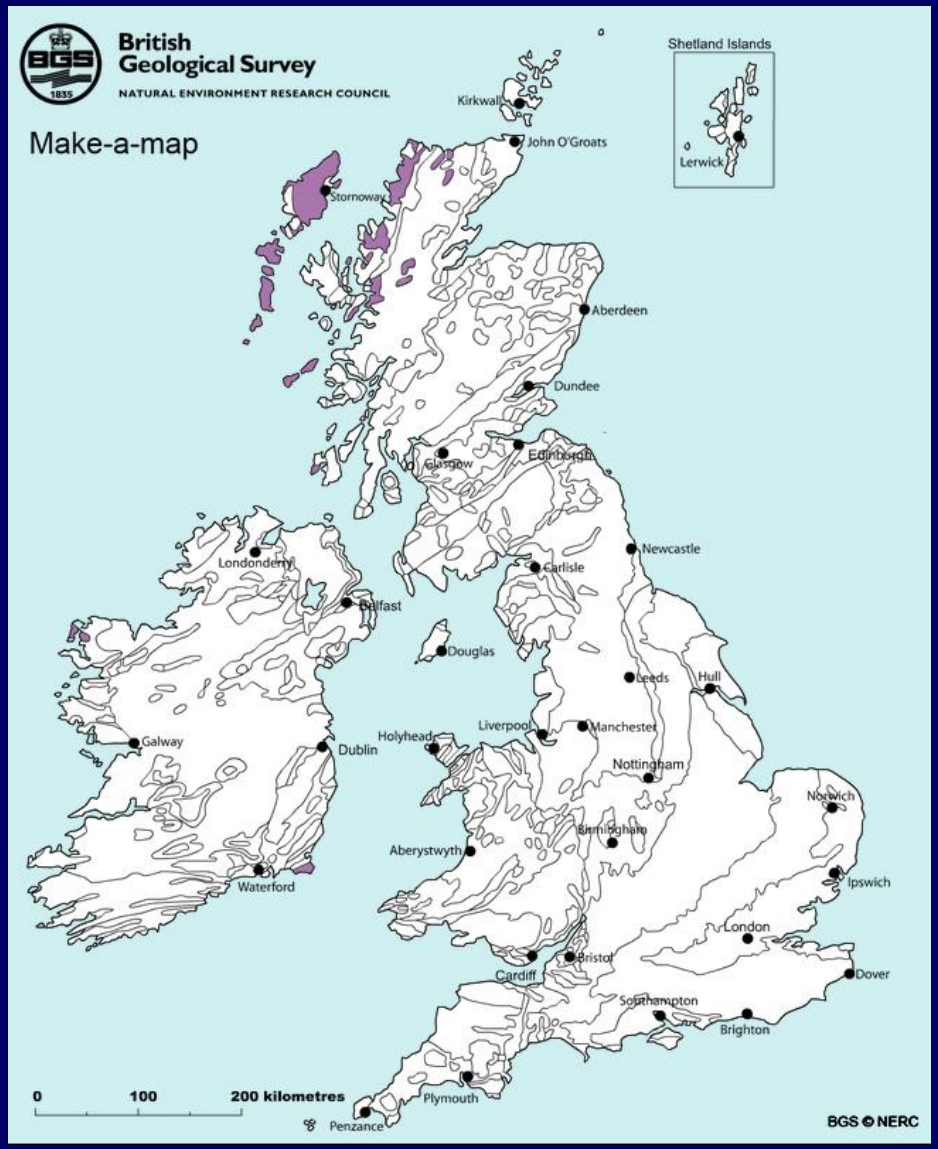


- High grade metamorphic rocks (gneiss)
- Intrusive felsic igneous rocks
- Tonalite – rich in quartz, feldspar (plagioclase and Na-rich alkali feldspar) and minor amphibole and pyroxene
- Some mafic rocks – remnants of early basaltic crust from the sea floor

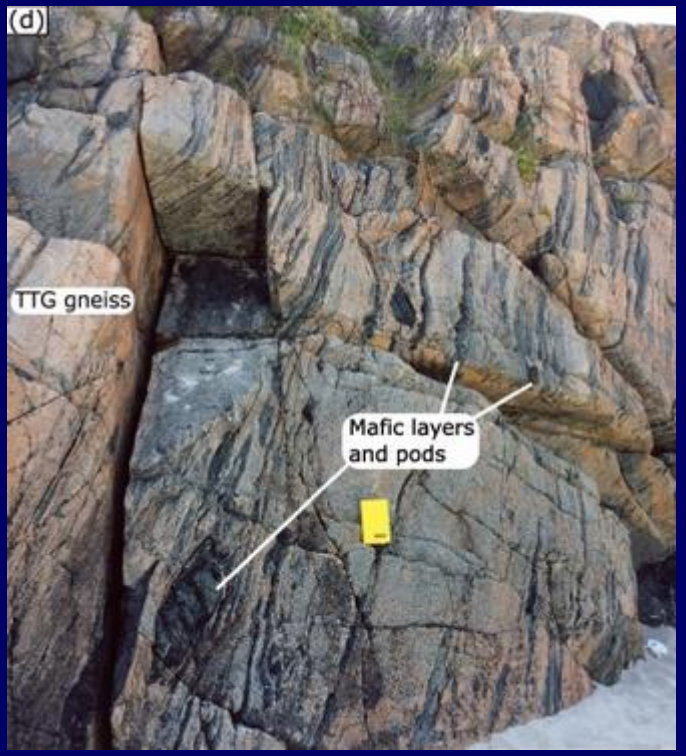


4030 - 3580 Ma Acasta Gneiss Complex in the Slave Craton, north-western Canada

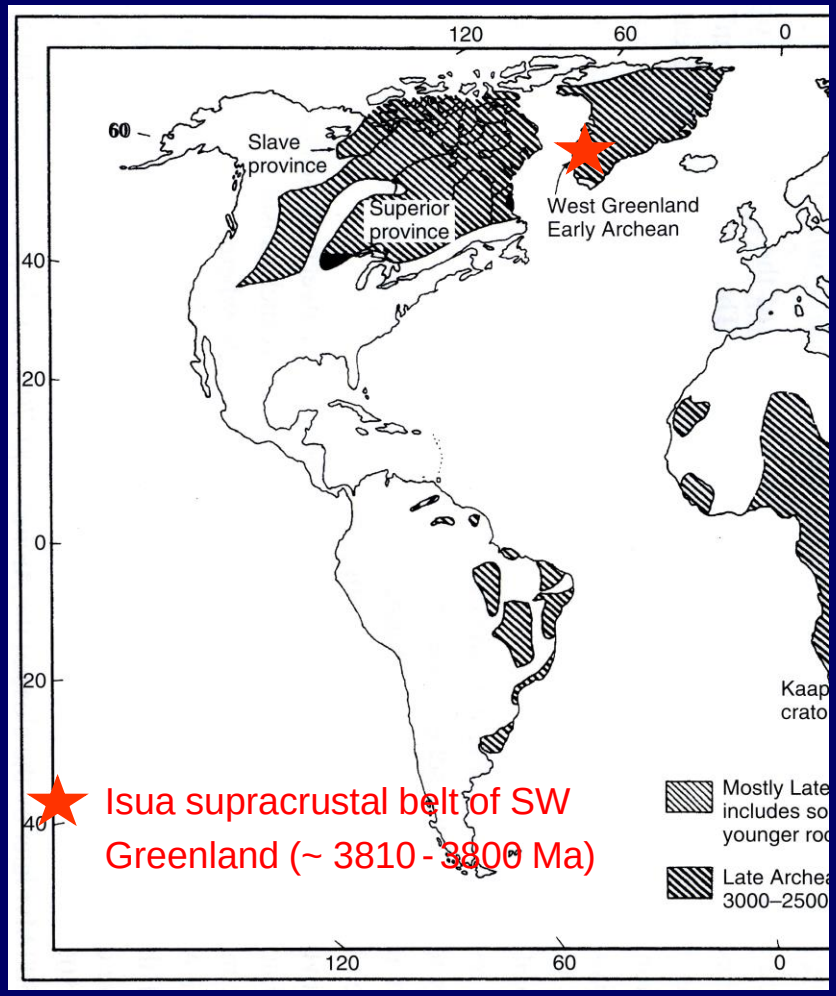
Archean rocks in the UK



- Oldest rocks in the UK are in NW Scotland
- Lewisian gneiss complex about 3Ga old
- High grade metamorphic rocks



The oldest sedimentary rocks



- Supracrustal – rocks formed at or near the Earth's surface lie on basement rocks formed at depth
- Stromatolites - microbial reefs created by cyanobacteria



Layered carbonate rocks in the Isua supracrustal belt. The rocks are strongly deformed and so the layering is not simple sedimentary layering

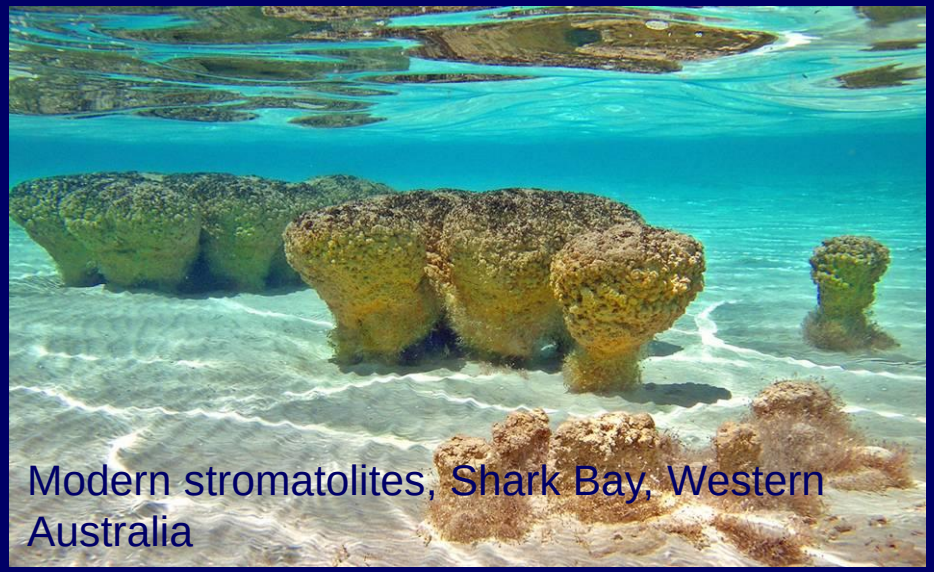


One of the two stromatolite controversial occurrences in the Isua belt (features are 1-4cm high)

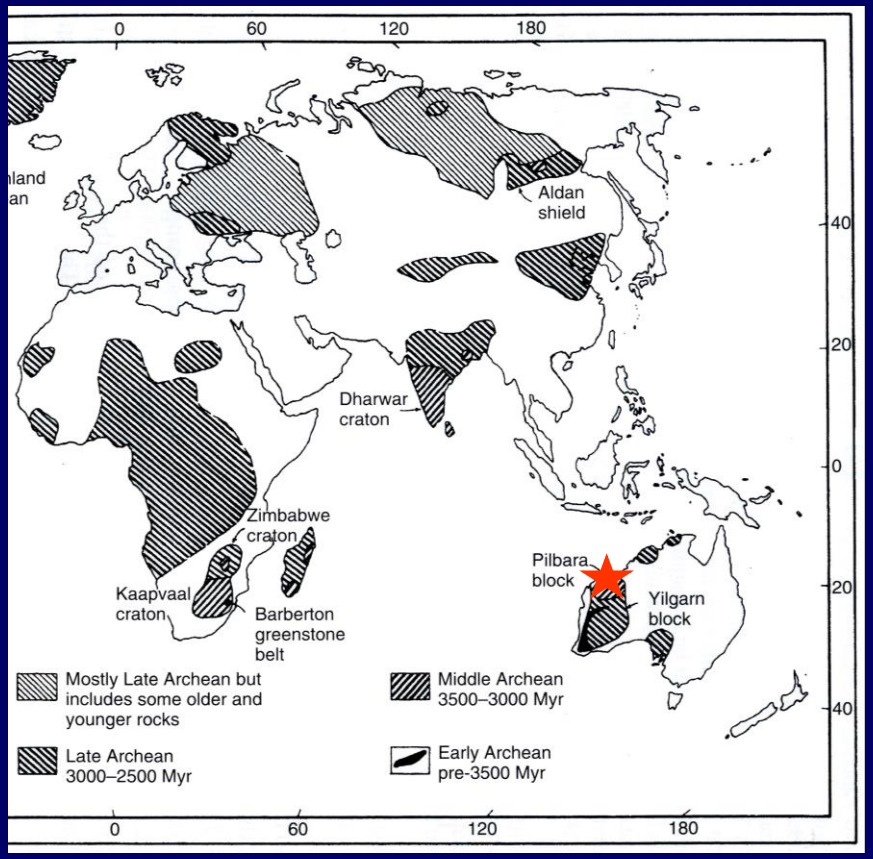
The first macroscopic fossils – stromatolites



Ancient stromatolite fossil;s preserved in the Dresser Formation, Western Australia

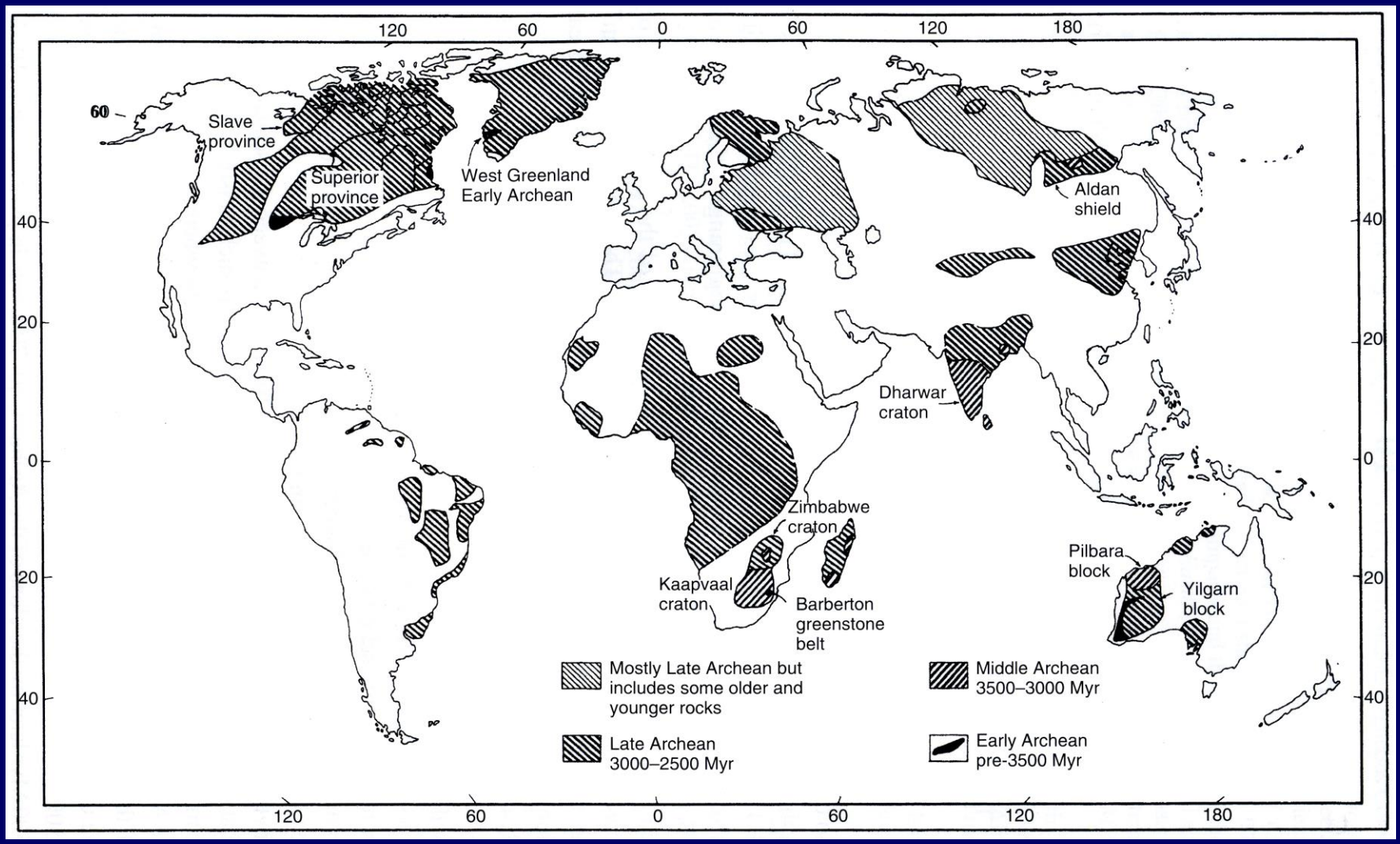


Modern stromatolites, Shark Bay, Western Australia



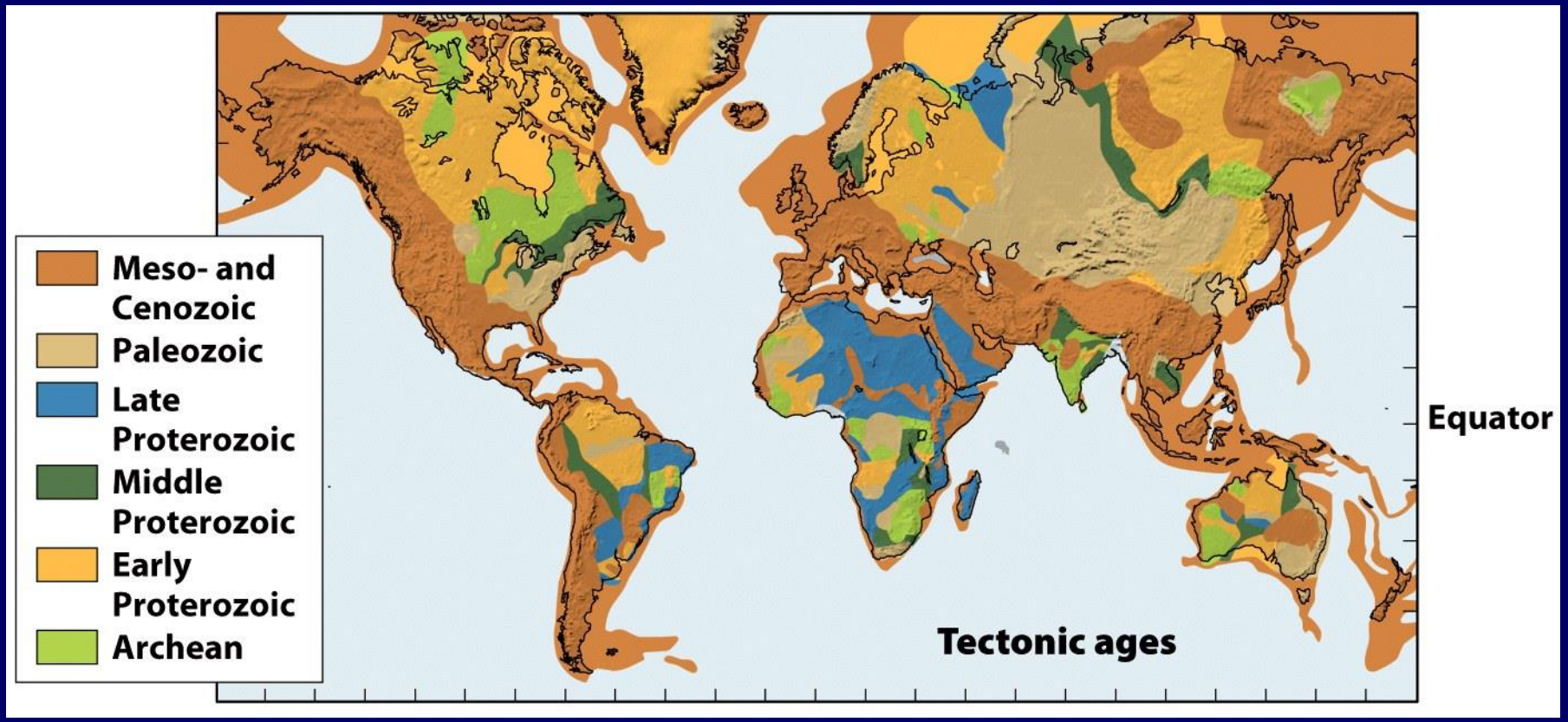
★ Dresser Formation, Western Australia – 3490 Ma

Distribution of Early, Mid and Late Archean cratons



Early Archean age rocks are rare, relative to rocks of Late Archaean age

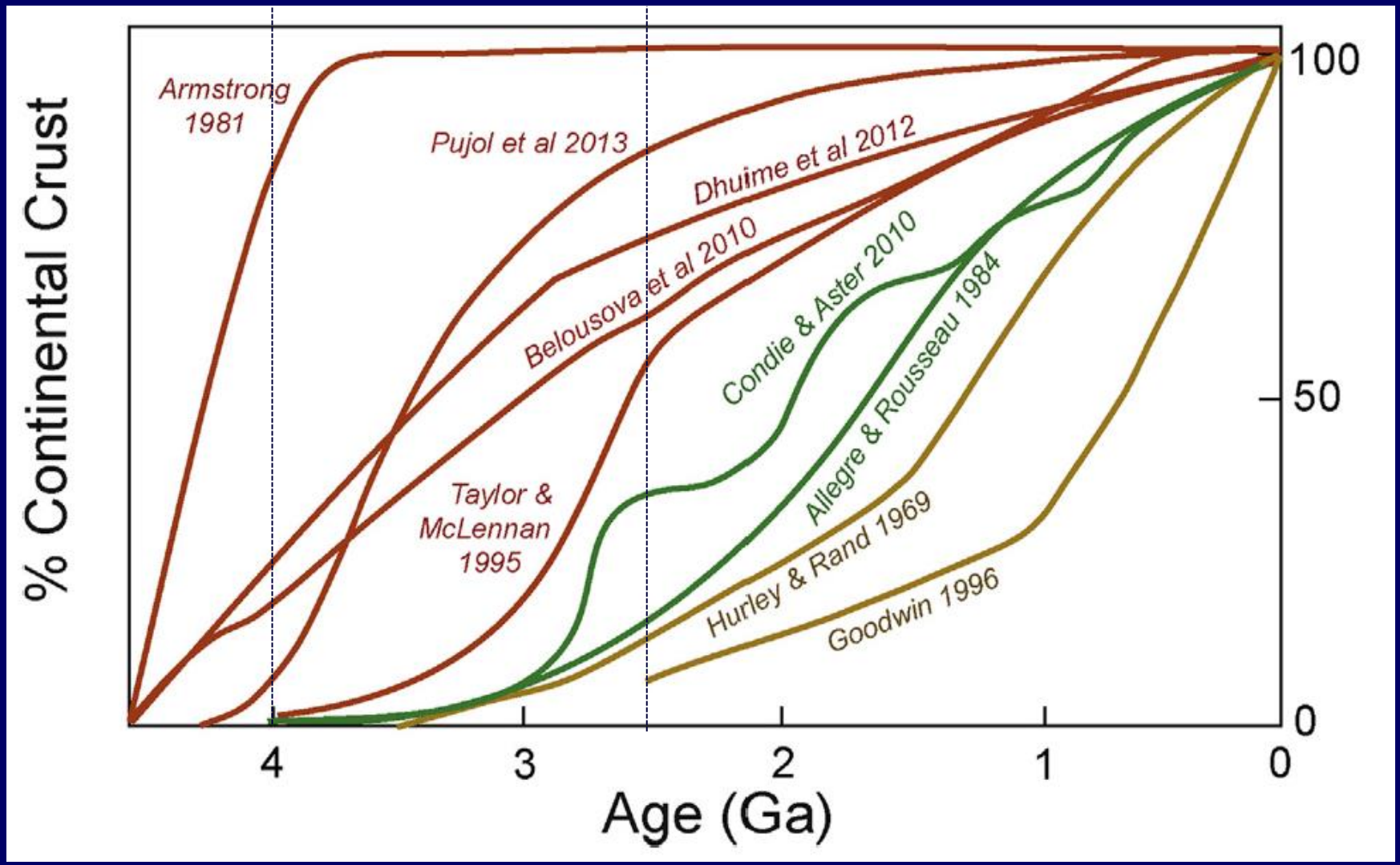
Continental growth rates through time



The aerial extent of continental crust of different ages today does not directly reflect the amount of crustal growth through time

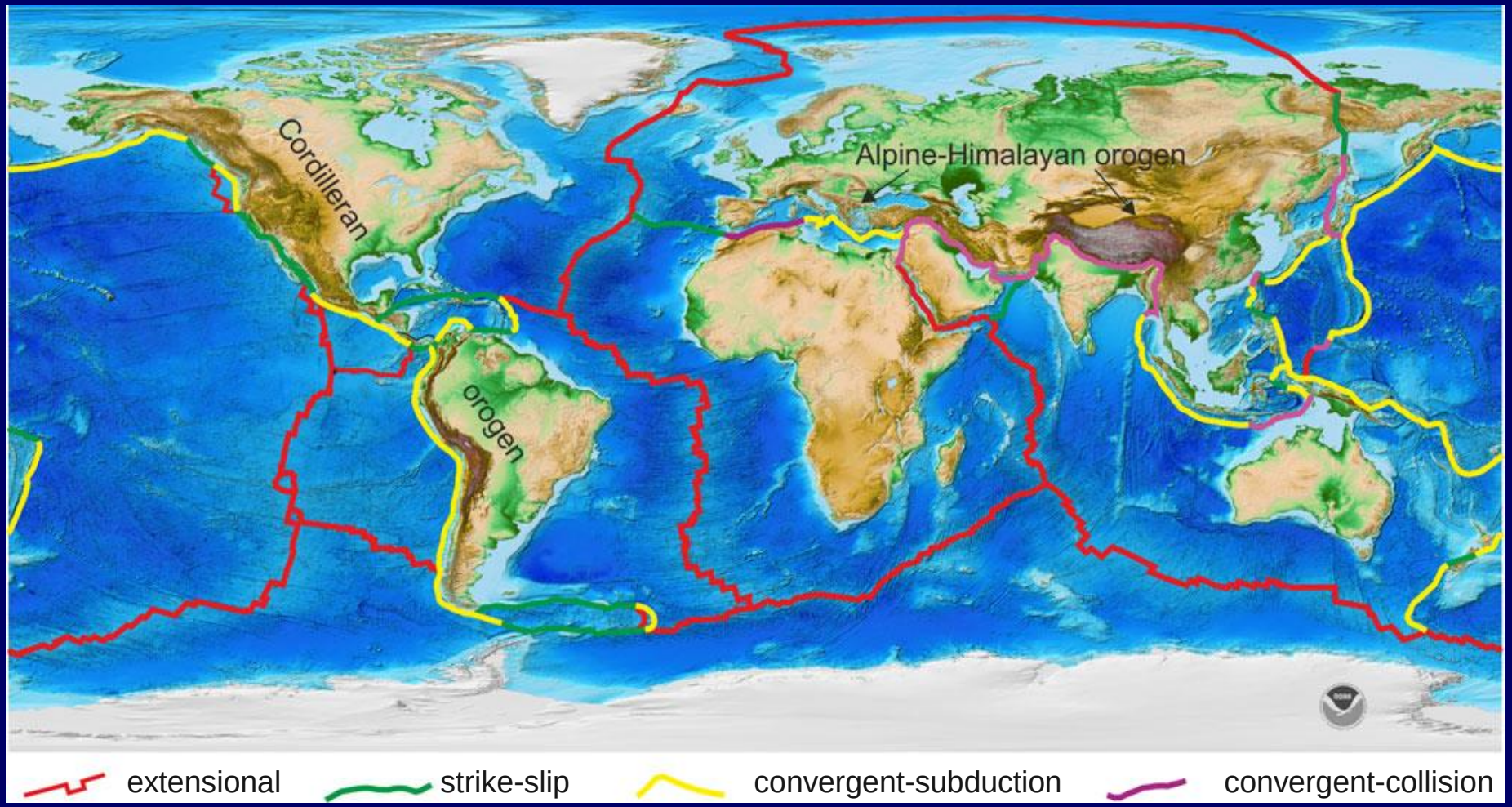
- Preservation: recycled into mantle when tectonic subduction began in the Archean or cycled by re-melting of existing crust (intra-crustal recycling)

Continental growth rates in the Archean



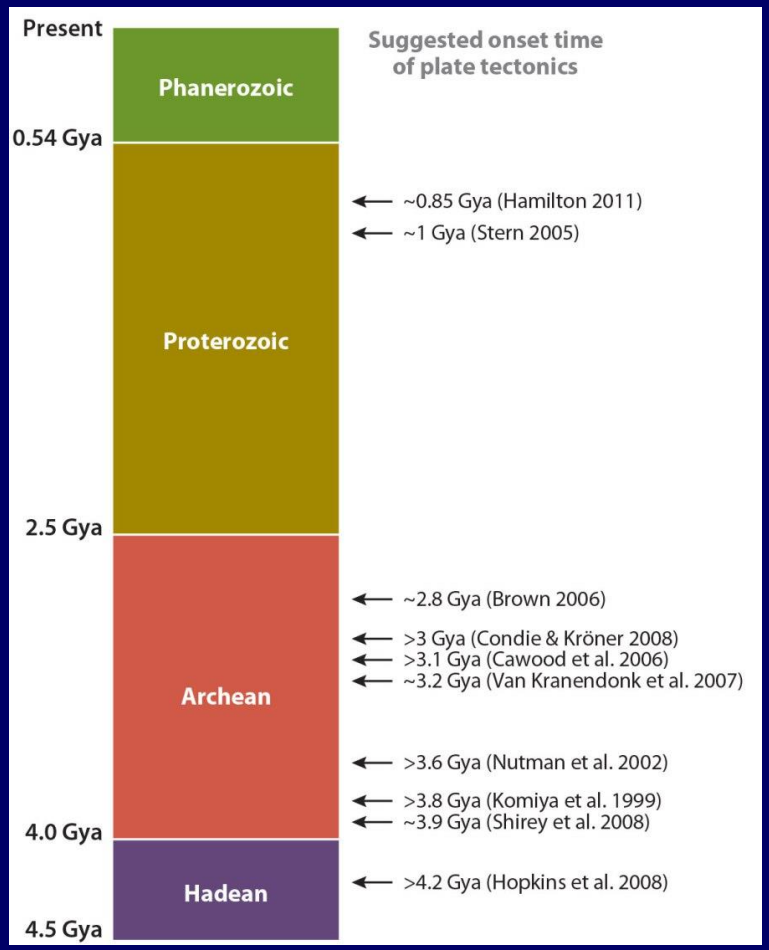
Selected crustal growth models

Present day linked plate boundaries



Present-day Earth global relief showing distribution of spreading, subduction, collisional and strike-slip boundaries

When did plate tectonics start?



Geologic timescale and diversity of opinions for the onset time of plate tectonics, from Korenaga (2013).

Large uncertainty and disagreement exists about when plate tectonics started

Indicators of plate tectonics:

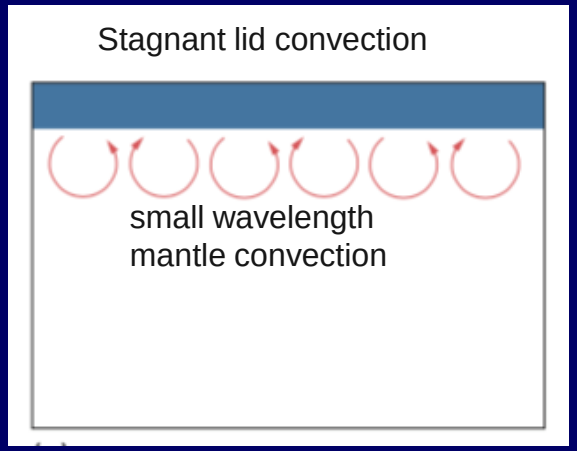
- rock sequences associated with plate tectonic settings
- geochemistry

Indicator	Widespread distribution (Ga)	First appearance (Ga)
UHP metamorphism	≤0.2	0.70
Paired metamorphic belts	≤2.8	3.2
Transcurrent faults and sutures	≤1.0	3.3 (?)
Collisional orogens	≤2.0	2.7
Accretionary orogens	≤2.0	4.0-3.8 (?)
Lateral plate motion	≤2.0	2.6-2.2
Geochemistry	≤2.0	3.8
Isotopes	≤2.2	≤3.0
Cratons	≤2.7	3.0
Dipping reflectors in lower crust	≤2.7	(?)

UHP, ultra-high pressure metamorphism.

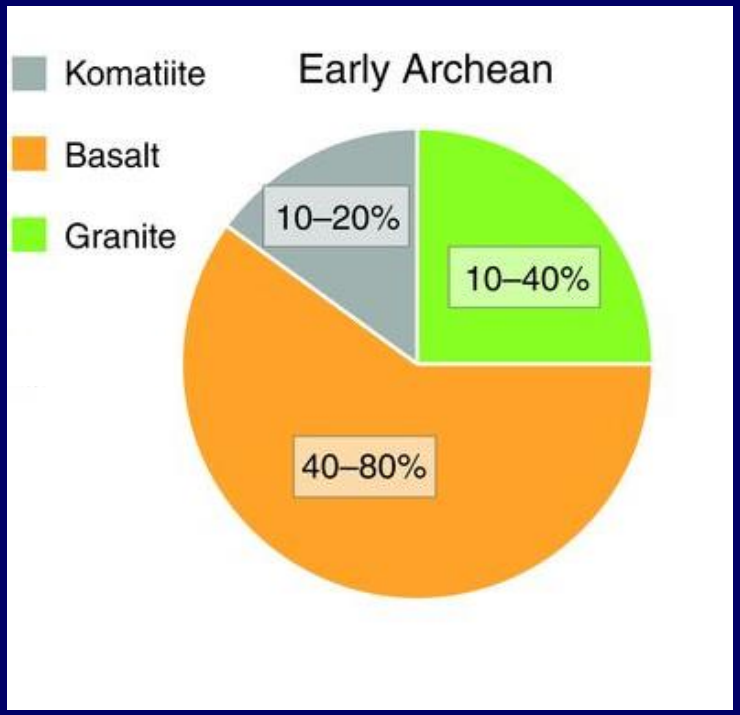
What came before plate tectonics?

Pre-plate tectonic model for the early Earth



- Stagnant-lid convection – an immobile, conductively-cooled lid, where underlying upper mantle develops unsteady convection cells .
- Planets in this regime are often referred to as single-lid or one-plate planets which is probably a better description.

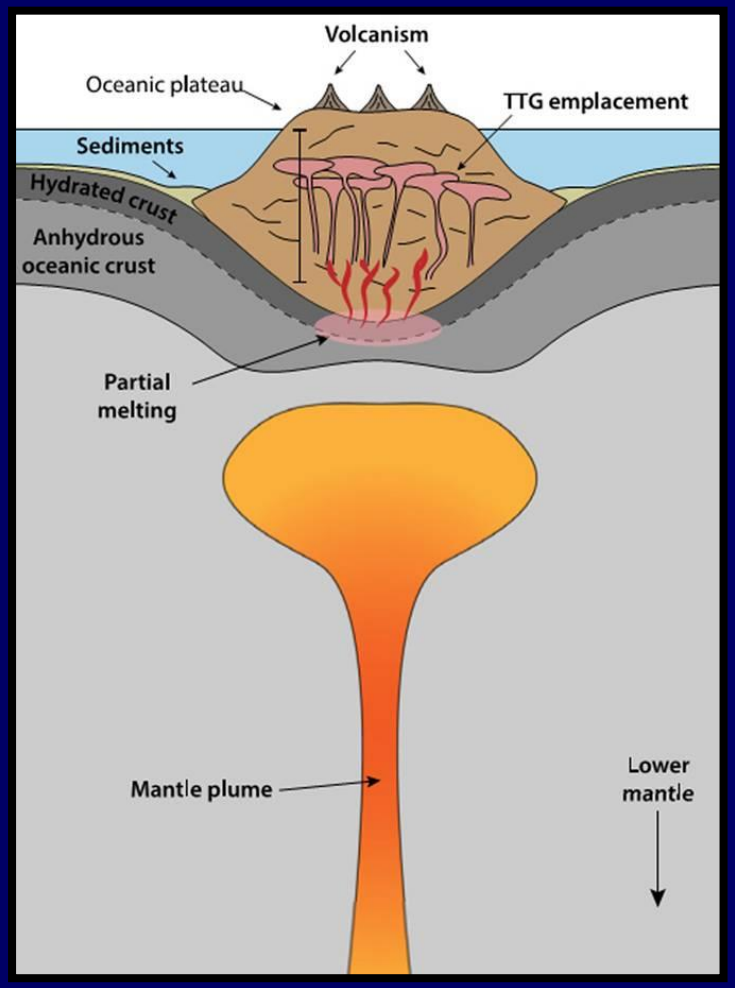




The three rock types that dominated the early continental crust

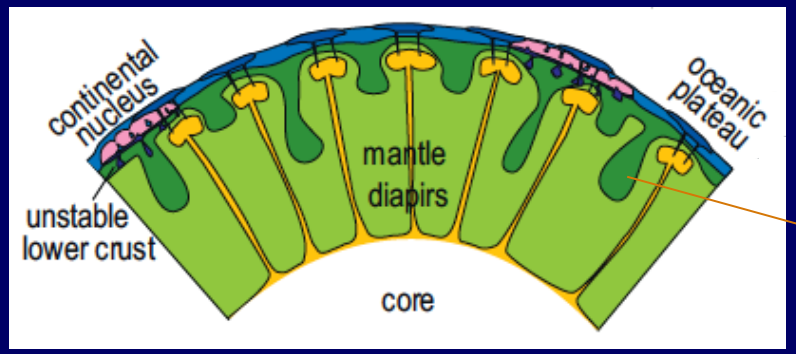
Komatiites are ultramafic (v. high Mg content) rocks common in the Archaean formed in a hot mantle

Partial melting and protocontinent development



- No deep subduction; mantle circulation bottom-up

Pre-plate tectonics

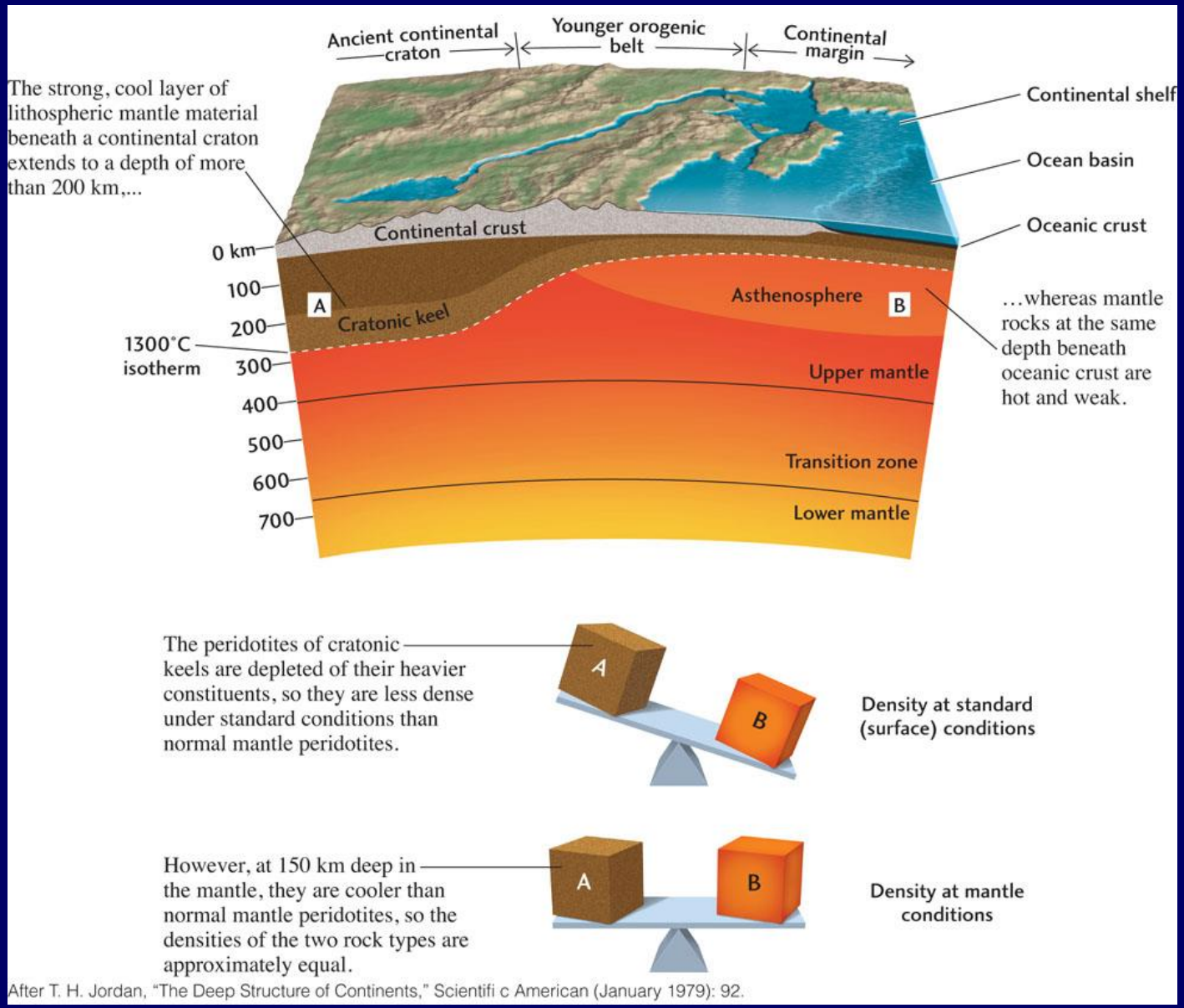


Deep mantle and plume driven bottom-up circulation

Stagnant lid tectonics

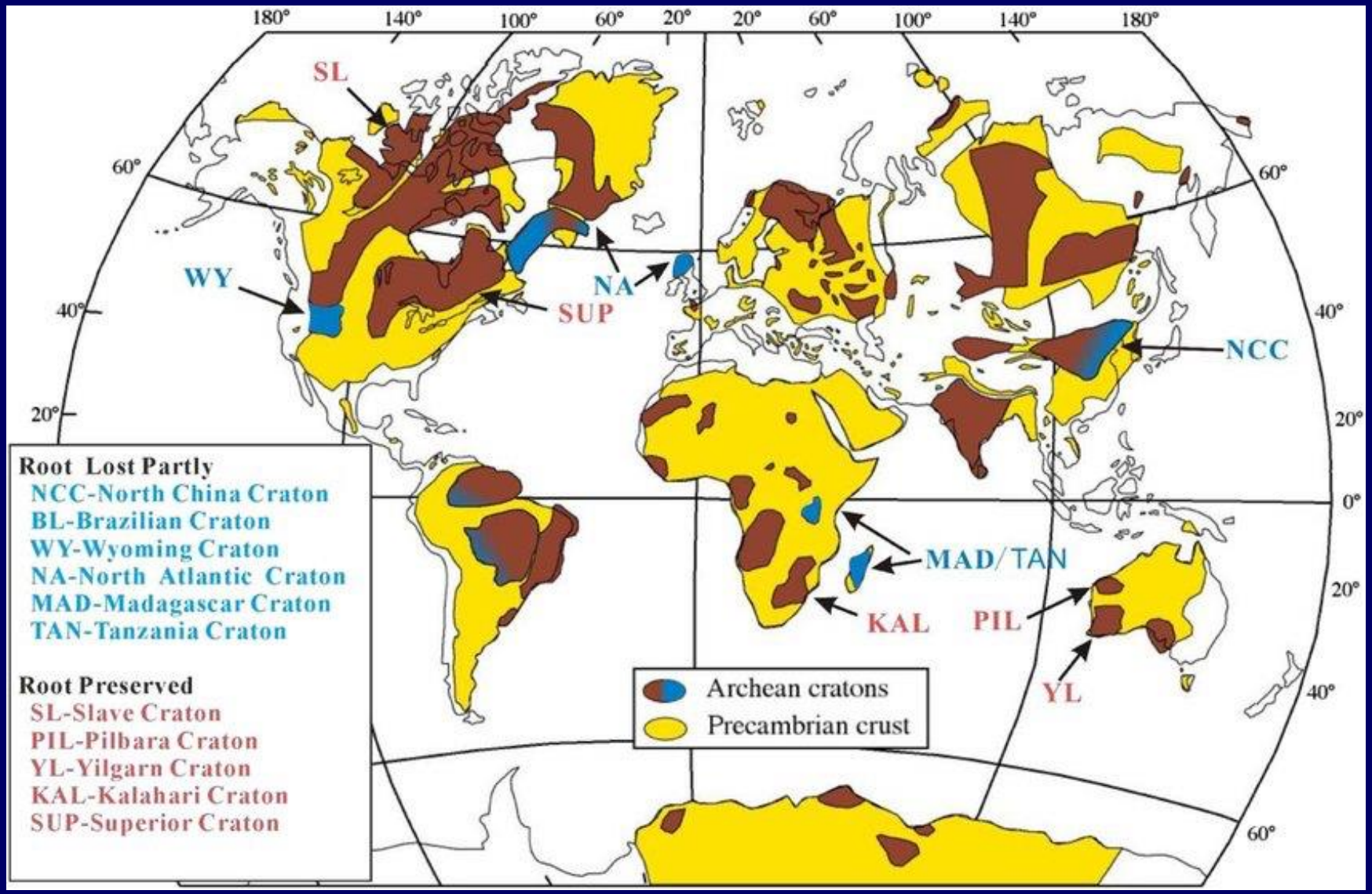
- Cooling by rising mantle plumes
- Numerous, larger and longer-lived than modern plumes
- Partial melting forms mafic lavas (incl. komatiites) and mafic intrusions
- Oceanic plateaus rise above sea level surface with crust thickened to 30-40 km
- Melt to give felsic intrusions – tonalite-trondhjemite-granodiorite (TTG) – quartz-feldspar rich granitic rock – formed first felsic continental crust and are major part of Archean cratons

Mid-Archean: Cratonic roots/keels form



After T. H. Jordan, "The Deep Structure of Continents," Scientifi c American (January 1979): 92.

Long-term stability of Archean cratons



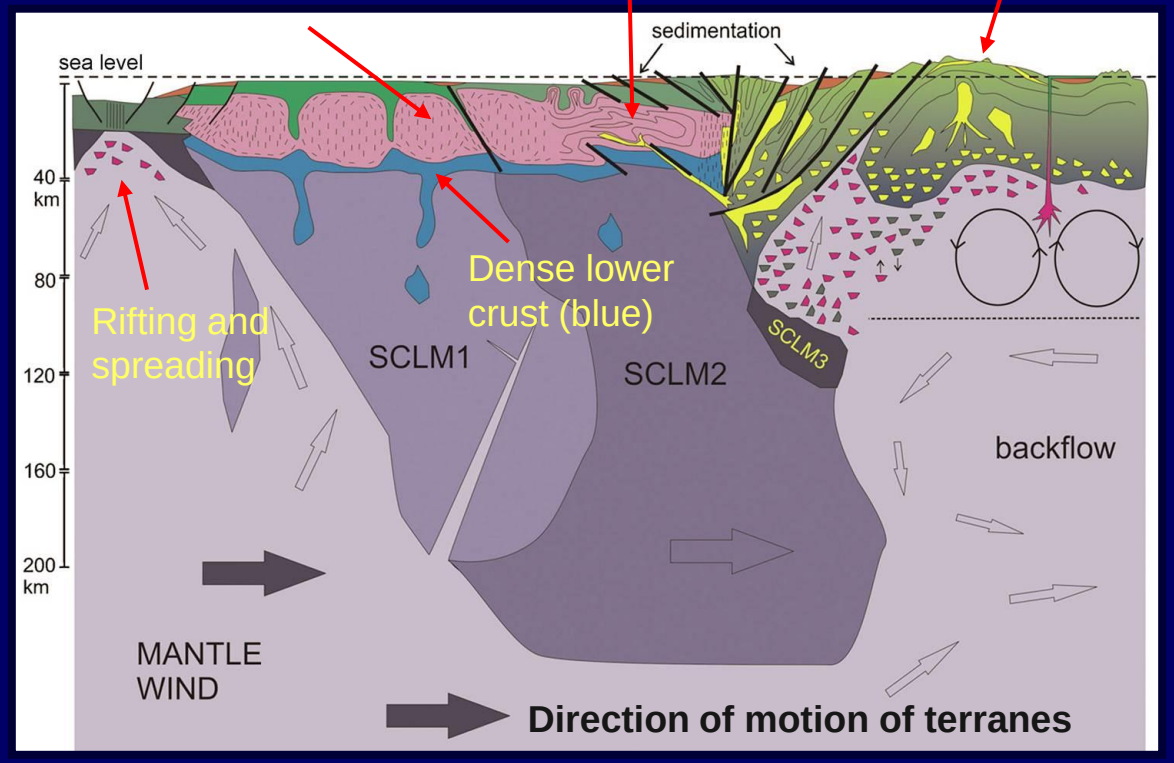
Mid-Archean continental drift, accretion & amalgamation

Archean lithosphere not rigid enough to survive deep subduction (no slab pull)
Outflow from upwelling mantle plumes pushed on the cratonic keels causing them to drift.

two amalgamated TTG (pink) terranes overlain by basalt (green) forming dome and keel structure

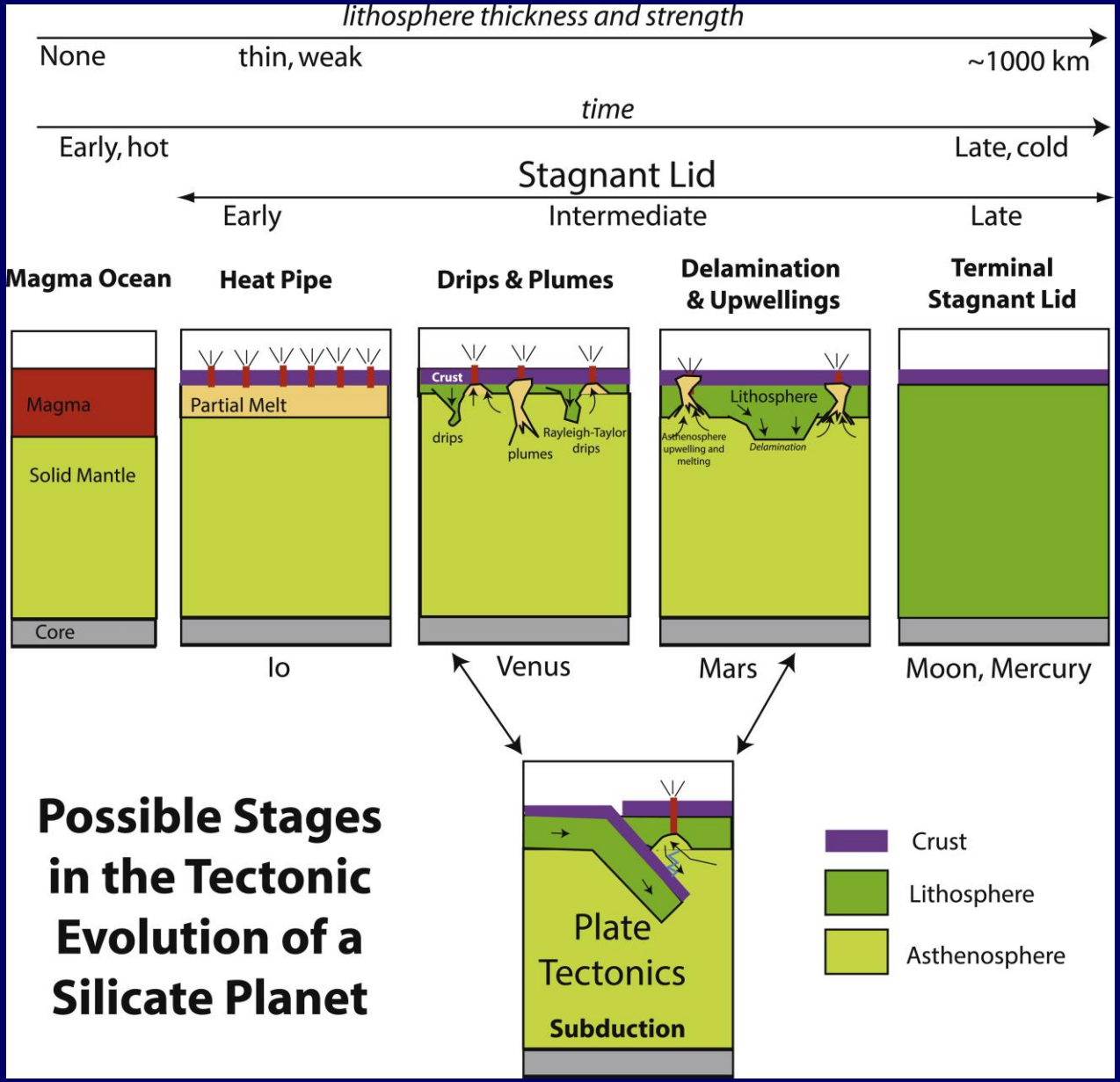
strongly deformed TTG gneisses

oceanic plateau crust (green) new TTG magmas (yellow)

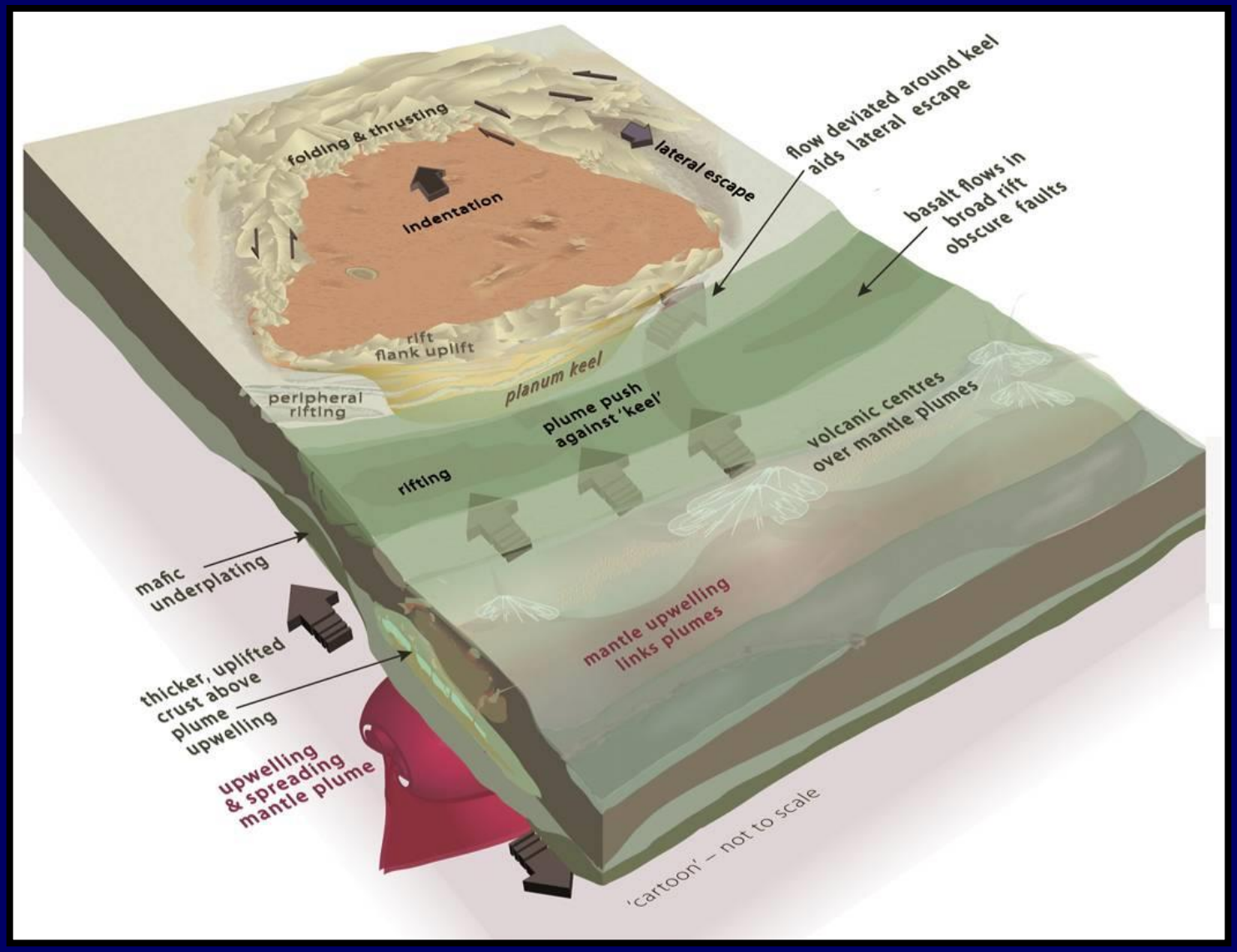


The formation of the SCLM provided something upon which horizontal mantle flow could act to drive lateral motion → terrane accretion and amalgamation

Tectonic styles in the solar system

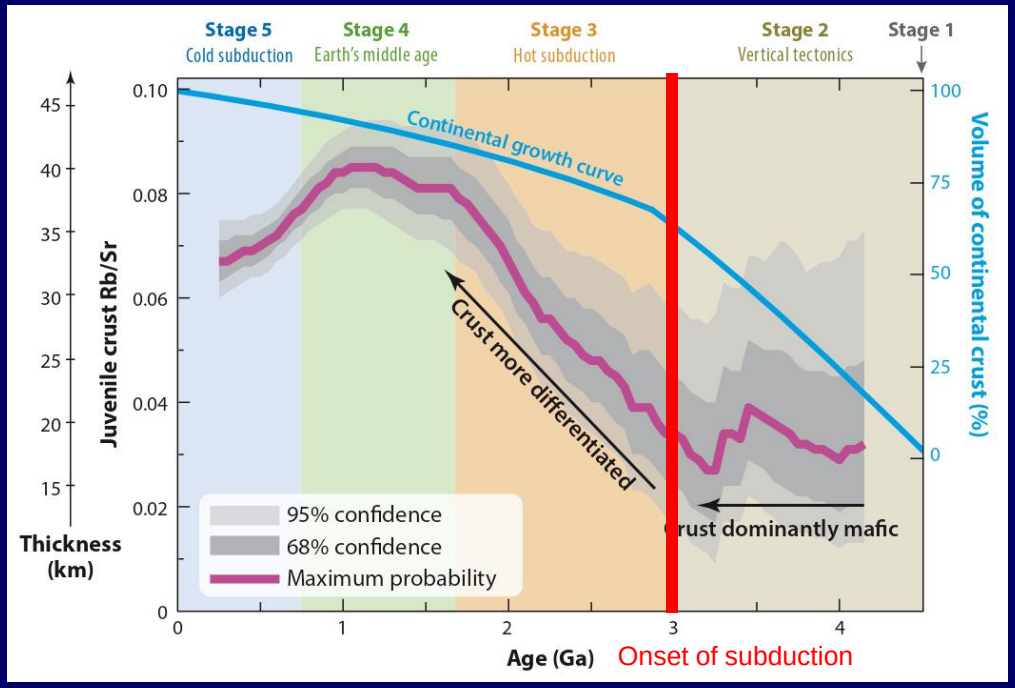


Venus: an analogue for the Archean Earth



Analogue of Archean tectonics based on a model of stagnant lid tectonics on Venus

Archean crustal growth (evidence for onset of subduction)

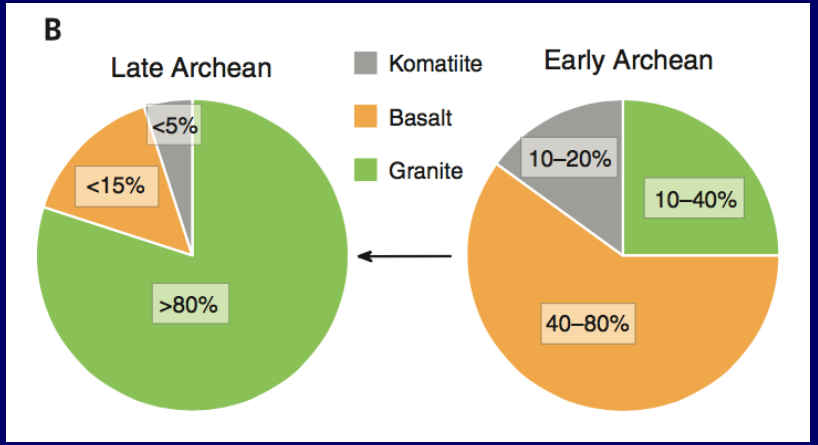


By 3000 Ma the mantle cooled sufficiently to develop a lithosphere of sufficient strength and rigidity for subduction to be possible

Key evidence for timing of onset of subduction around 3.0 Ga:

- slow down in rate continental crust formation (increase in rate of destruction)
- crustal recycling (Rb-Sr, Hf, Oxygen isotope ratios of magmatic zircons)
- volumetric increase in TTG granitoids
- relative shift from Na to K-rich granites
- change in compositional of continental crust from mafic to intermediate

The mafic upper continental crust in the early Archean was gradually replaced by a felsic upper continental crust and reached an average composition much like today around the Archean-Proterozoic boundary

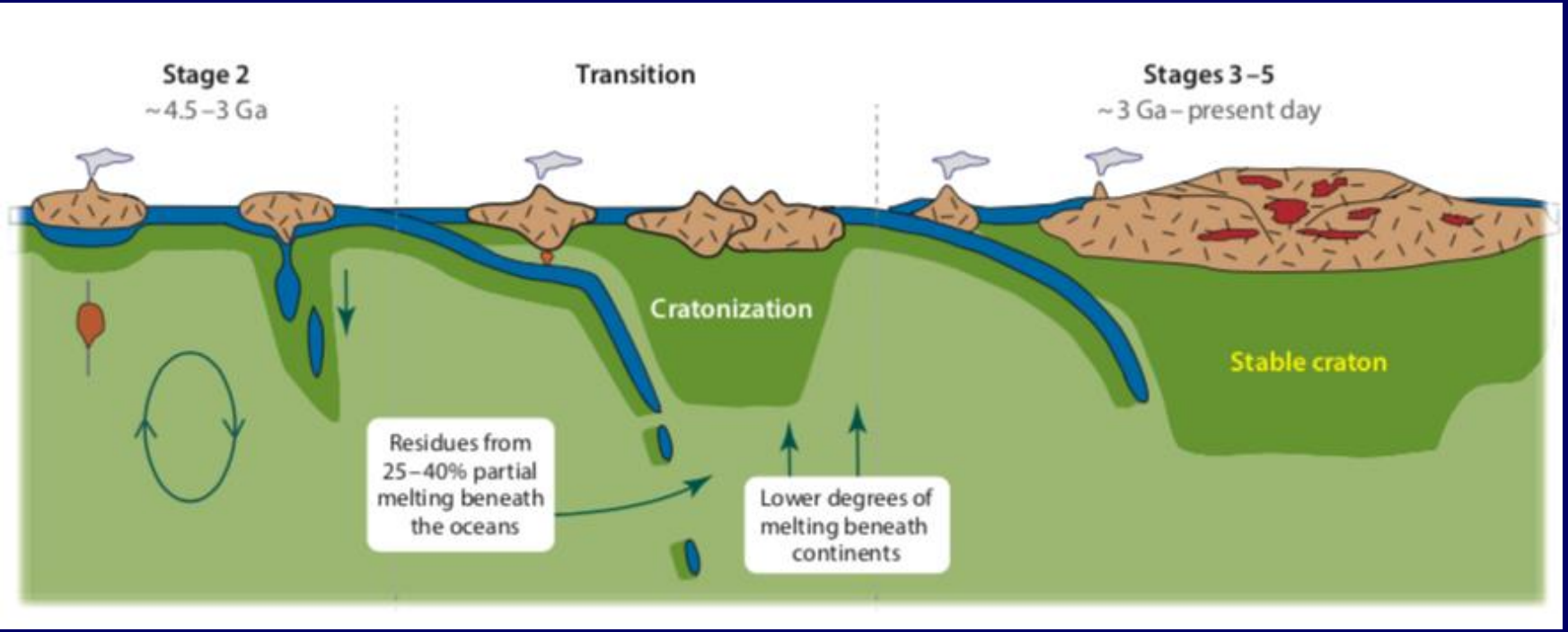


MgO content ~4 wt% (present day 2-3 wt%)

MgO content >11 wt%

The road to modern plate tectonics

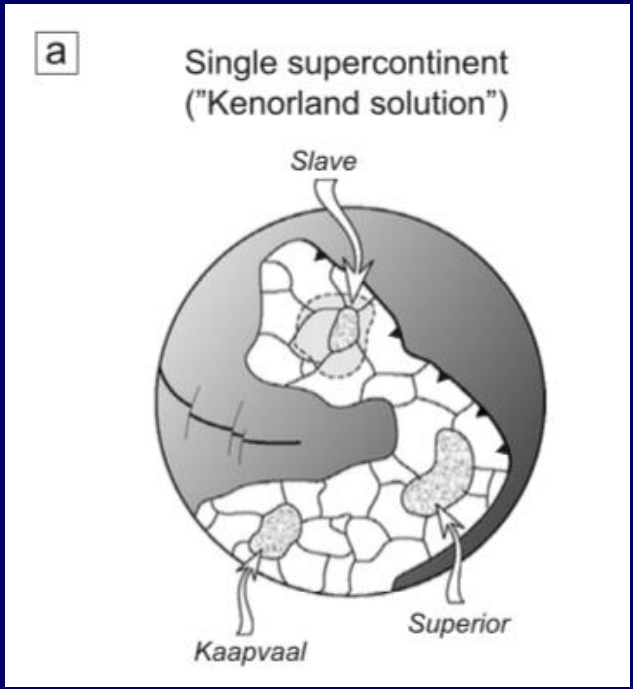
- in the Hadean / early Archean subduction was inhibited and magmatism was driven by rising mantle plumes that “percolated the lithosphere”
- as the mantle cooled, lithospheric strength and rigidity increased, subduction became possible
- early subduction was probably episodic and short-lived: weak slabs → early slab break-off → loss of slab-pull → subduction ceases
- considerable variability in the structural style of orogenic belts depending on the thermal state of the deforming units

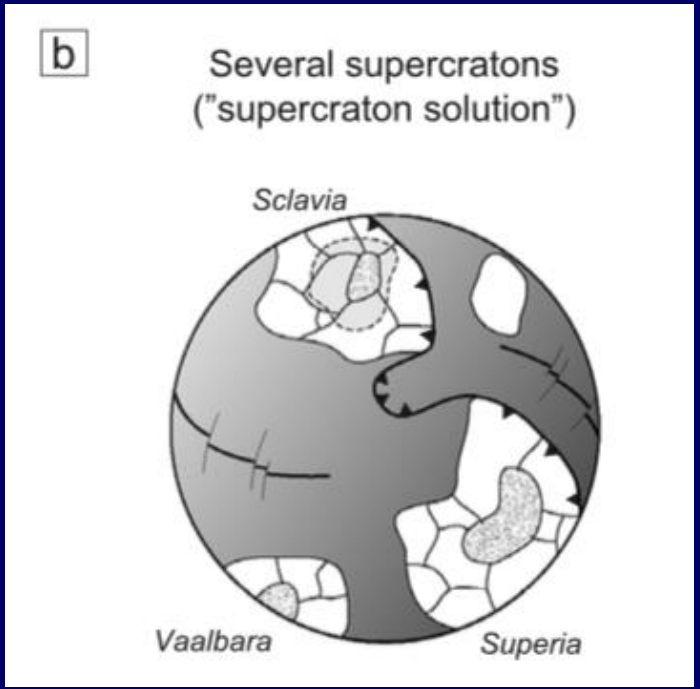


Subduction enhanced continental collisions → the creation of larger continents

Formation of the continents

In the late Archean the continents were perhaps assembled into a supercontinent (e.g. Kenorland 2750-2500 Ma)





Alternative ideas:

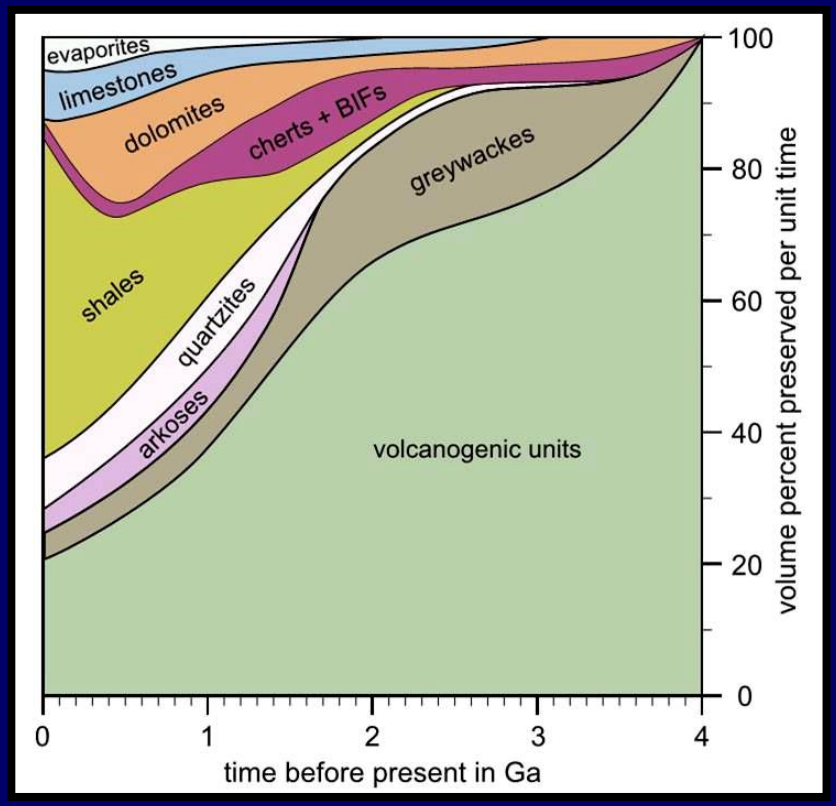
- Based on distinctive geology: 3 separate continents (Vaalbara, Superia and Nunavutia)
- Based on palaeomagnetic grounds – single stable supercontinent (Protopangaea) 2700-600 Ma

Deposition in the Archean

- Development of deep buoyant cratonic roots, stiffened the continents allowing them to emerge above seal level and support mountain chains
- Erosion of the mountains delivered large volumes of clastic sediments to oceans
- The continents provided a basement for sedimentary platform successions several kilometres thick and shallow, stable continental shelves:

Volcanogenic units: volcanicalstics and lavas
Greywackes: mixture of sand and mud eroded from the volcanic areas
Quartzites: texturally mature quartz rich sandstones that have been recrystallised
Carbonates: predominantly dolomitic (chemogenic rocks) but later limestone (biogenic)

Chert (amorphous silica) and Banded Iron Formations (BIFs; ironstones)



Iron Minerals in Sediments

Many sedimentary rocks contain a few percent of iron giving the yellow, brown and red colours that are so common



Haematite (Fe_2O_3) is the most common oxide

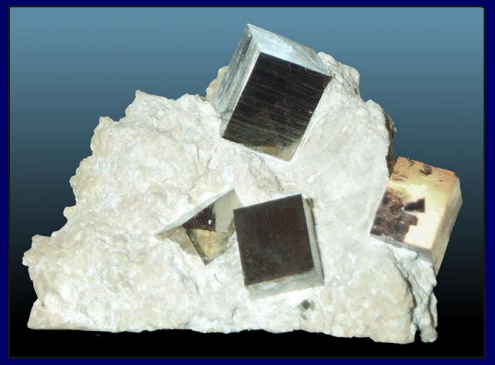
Other common iron-bearing minerals in sediments



Magnetite (Fe_3O_4)
(another iron oxide)



Goethite (FeO.OH)
(iron hydroxide)



Pyrite (FeS_2)
(iron sulphide)



Greenalite ($Fe_{2-3}(Si_2O_5)(OH_4)$)
(iron-rich silicate mineral)



Siderite ($FeCO_3$)
(iron carbonate- diagenetic)

What is an ironstone?



Pyrite-rich mudstone



Oolitic ironstone



Ironstone concretion



Banded iron formation

Laterites

- Laterites are iron oxide rich soils ("limonite" mixed Fe-hydroxides)
- Formed by intense chemical weathering of iron-rich igneous rocks



Occur in humid tropics where all soluble components have been carried away leaving residual iron oxide rich deposit

Found today e.g. Brazil, West Africa

Banded Iron Formations (BIFs)



Global distribution of economic BIFs



Precambrian Banded ironstone. Darker layers are rich in iron oxides, lighter layers are mostly chert (amorphous silica).

No equivalent forming today

- All Precambrian in age, predominantly between 3500 and 1900Ma

Origin not fully understood – silica-rich Archean ocean, periodic ppt of Fe

Origin of BIFs: Source of soluble reduced iron (Fe^{2+})

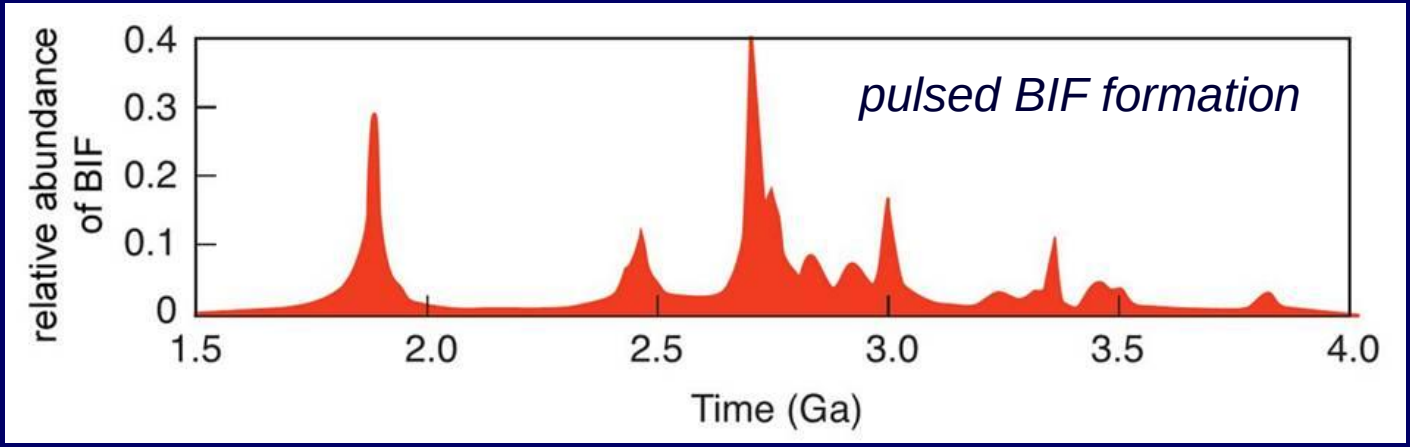
- Chert and iron minerals were precipitated on widespread continental shelves or shallow basins.
- The lack of obvious current or wave generated structures suggest they were deposited below the wave base at >200 m depth.
- The thick accumulations of BIFs require huge quantities of dissolved Fe2+ to be present in the oceans.



- There are two ways to achieve this:
  - oxygen-free ocean, or
  - stratified oceans with anoxic lower part containing Fe in solution

Origin of BIFs: Source of soluble reduced iron (Fe^{2+})

(1) Iron source: hydrothermal vents on sea floor (volcanic)



The primary source of iron was submarine volcanic activity and so the pulses reflect periods of increased activity

(2) Iron source: Deep weathering of continental rocks

- Formed at a time when there was little or no oxygen in the atmosphere, and higher CO_2 content
- higher partial pressure of CO_2 lowered pH of surface waters and led to greater efficiency in iron leaching and transportation
- Oxygen-poor atmosphere means Fe^{2+} from continental weathering transported to oceans

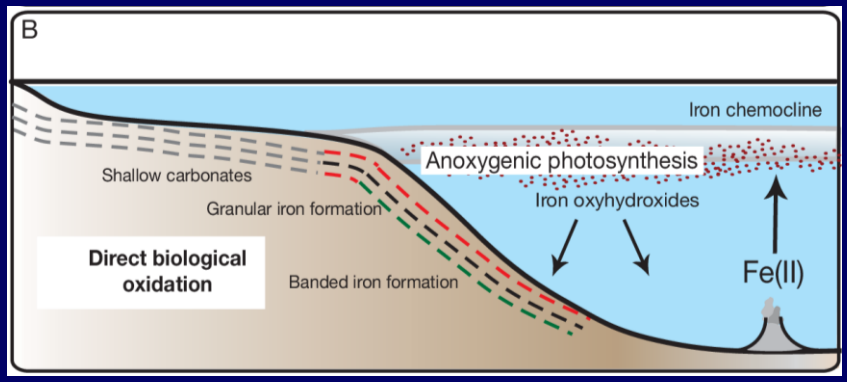
Microbial Mechanisms for oxidation of Fe^{2+}

Microbial mechanisms for oxidation of Fe (ie converting soluble Fe^{2+} to insoluble Fe^{3+} compounds):

Both involve photosynthesis: process that converts light energy into chemical energy to carry out metabolic processes

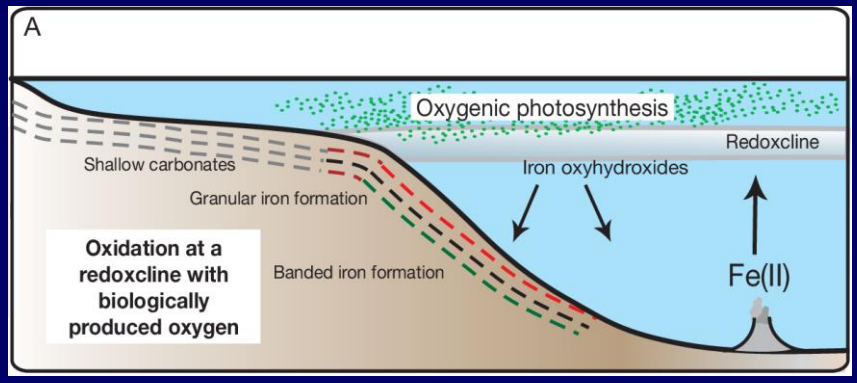
Anoxygenic Fe^{2+} oxidizing photosynthesis

- bacterial photosynthesis carried out under anaerobic conditions
- $4Fe^{2+}_{(aq)} + CO_2 + 4H^+ + light = CH_2O + 4Fe^{3+}_{(aq)} + H_2O$
- direct oxidation of Fe^{2+}: ferrous ion (Fe^{2+}) used as an electron donor
- no oxygen produced

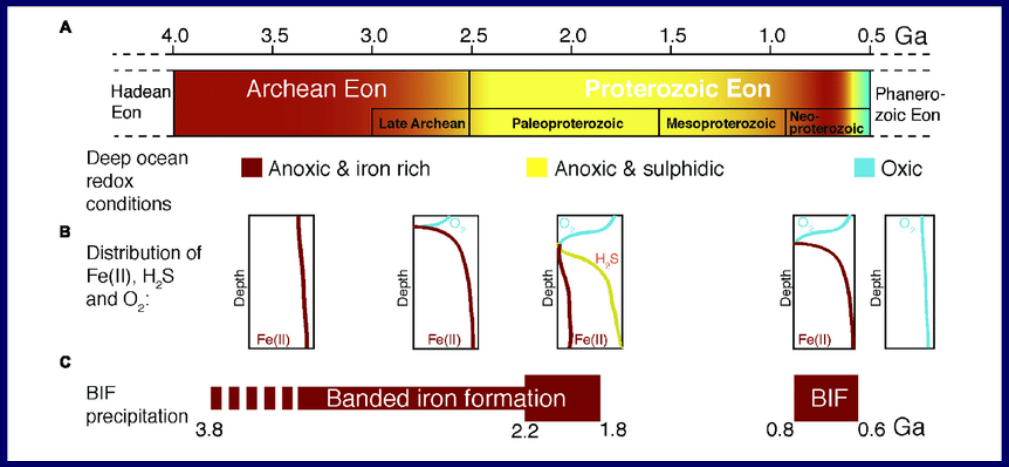


Cyanobacterially driven O_2 production

- $H_2O + CO_2 + light = CH_2O + O_2$
- production of O_2 is a 'waste product' of oxygenic photosynthesis by cyanobacteria
- abiotic process: independently, oxygen combines with Fe^{2+} in the sea water to produce ferric oxide/hydroxide precipitates

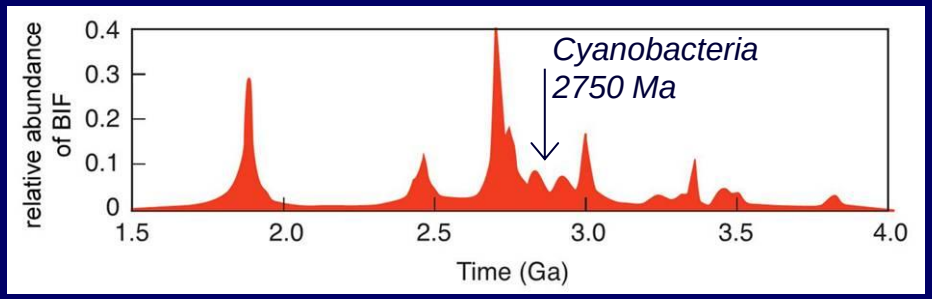


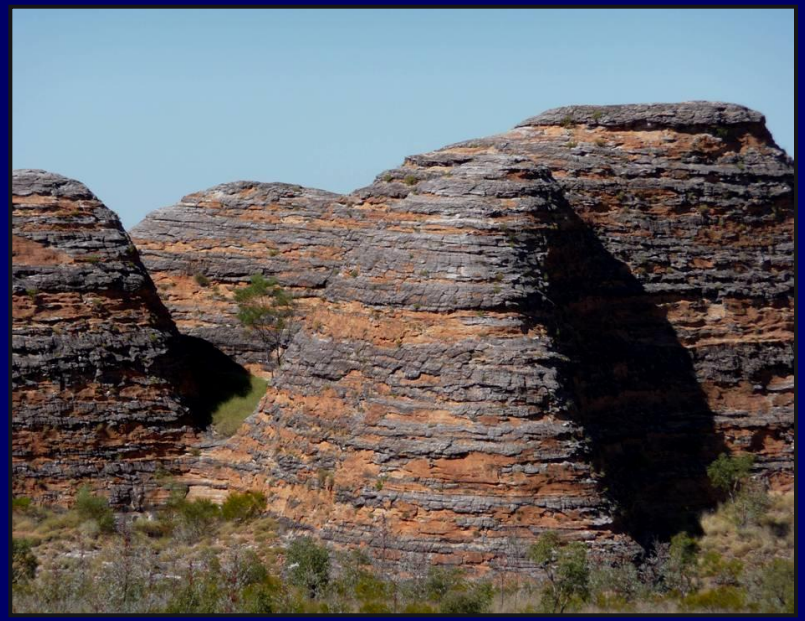
Formation of BIFs





- Early in Archean, Fe^{2+} oxidation in anoxic oceans by anoxygenic phototrophs
- After emergence of cyanobacteria, abiotic oxidation of Fe^{2+} dominant
- oxygen levels in shallow water areas of oceans began to rise
- iron oxidised to form insoluble Fe^{3+} compounds and precipitated on the sea floor as hematite Fe_2O_3

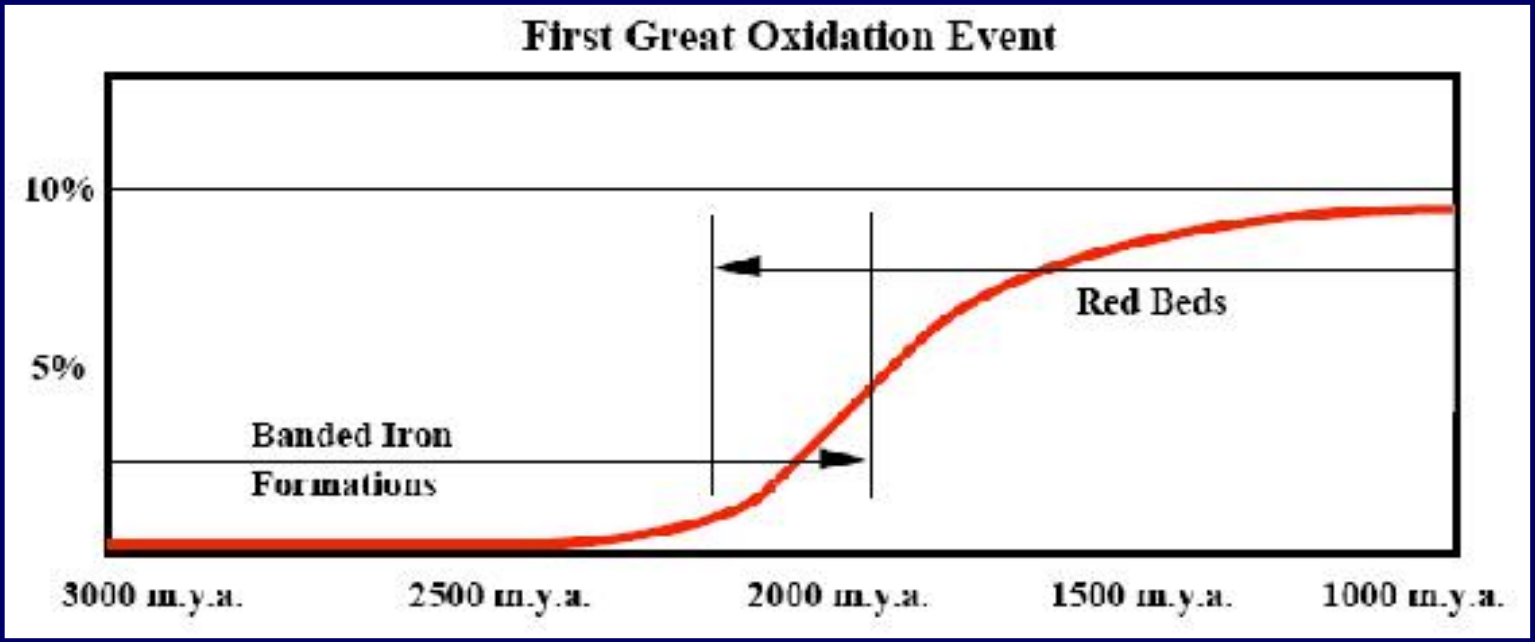




The end of BIF production and changes to atmospheric oxygen

Oxidation of iron consumed all the oxygen – it was not until the iron in the ocean waters was ‘used up’ that oxygen levels in the atmosphere started to increase (Great Oxidation Event)

After that, iron oxidised on the continents



Economic significance today



(Photo: Reuters/Esther Thomas /)

Summary of key events in the Archean

- Period of voluminous increase in addition of new continental crust
- 65-70% of current continental crust in place by 3.0 Ga
- Early crust mafic with felsic intrusions
- As Earth cooled, the lithosphere thickened and more rigid, allowing episodic, shallow subduction to begin around 3.0 Ga
- Subduction enhanced collisions leading to the formation of larger continents
- The continents became subaerial and mountains formed and were eroded
- Thick deposits of volcanogenic, chemogenic and clastic sediments deposited in basins
- Interactions between tectonic processes (volcanism, creation and uplift of the continents), life and the atmosphere resulted in the precipitation of thick BIFs which are the major source of iron today

Learning objectives revision

Now we have completed this class you should be able to:

- identify important Archean age rock localities on a map of the world
- chart changes in continental growth rates and explain how most continental crust today came into existence during the Archaean
- describe how the tectonic style of the Earth evolved during the Archaean and how the first stable continents formed
- list key differences in tectonic style to today
- list the main rock types present in the Archaean
- explain how the evolution of the oceans, atmosphere and life led to the deposition of Banded Iron Formations and elucidate why they are not forming today